

GAUTHAM MURALI

MCA GRADUATE | PYTHON | AGENTIC AI & RAG SYSTEMS | DATA ANALYSIS

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SUMMARY

MCA graduate with hands-on, project-based experience building agentic AI applications in Python using the LangChain ecosystem (LangGraph), Google Gemini API, and Ollama/Mistral. Skilled in designing multi-agent, stateful workflows with multi-step reasoning and tool usage, Retrieval-Augmented Generation (RAG) pipelines using vector databases and text embeddings, and prompt engineering for reliable model outputs. Strong fundamentals in Python software engineering, data structures, and version control, with a foundation for extending into cloud-native and enterprise-scale GenAI systems.

SKILLS & TOOLS

Programming: Python, JavaScript, HTML/CSS, MySQL, SQLite

GenAI & LLM Ecosystem: LangChain Ecosystem (LangGraph), Agentic AI Workflows, Google Gemini API, Ollama (Mistral), Prompt & Context Engineering, Retrieval-Augmented Generation (RAG), Vector Databases (Qdrant), Text Embeddings, Transformer-based Models

Frameworks & Tools: Django, Pydantic, PySide6, NumPy, Pandas, Scikit-learn, Git (Version Control), VS Code, Jupyter Notebook

Architecture & Concepts: Multi-Agent Systems, Multi-Step Reasoning, Human-in-the-Loop (HITL), Stateful Workflows, Semantic Search, Data Structures

PROJECTS

FlexRoute — Gym & Fitness Agentic AI App

- Designed and implemented an agentic AI workflow using LangGraph (LangChain ecosystem) with a deterministic semantic router, enabling multi-step reasoning and dynamic intent classification.
- Built Human-in-the-Loop (HITL) checkpoints to validate structured JSON context before syncing state to SQLite, improving reliability and safety of model outputs.
- Engineered a hybrid LLM routing engine that balances API cost and latency by dynamically toggling between local Mistral and cloud-based Gemini APIs.

AI Research Studio — Autonomous Agentic Engineering

- Built a multimodal, autonomous AI workbench in Python using the Google GenAI SDK, integrating LLM APIs into an agentic application with tool usage and autonomous decision-making.
- Implemented a ReAct execution loop enabling multi-step reasoning to decompose complex objectives into sequential tasks.
- Applied prompt engineering strategies and dynamic web-grounding to improve output reliability and reduce model hallucinations.
- Optimized UI responsiveness using a non-blocking, event-driven architecture for real-time model interaction.

Fake News Detection System

- Developed and trained a supervised NLP classification model in Python (Scikit-learn, Pandas) to detect fake news from text input and YouTube video content.
- Built the end-to-end data preprocessing and feature-extraction pipeline, applying core NLP and text-embedding techniques for model training.

EDUCATION

Master of Computer Applications (MCA)

APJ Abdul Kalam Technological University

2023 – 2025

Bachelor of Computer Applications (BCA)

University of Kerala

2019 – 2022

CERTIFICATIONS

- IBM AI Developer Professional Certificate — 2026
- NPTEL – Internet of Things (IIT Kharagpur) — 2023